

Artifact-backed qualitative diagnostic cases



Clean success
Detector-bridge sweep: PickCube
GroundingDINO / Crop-median depth
same shape distractor L0 seed 5
success=True error=0.025 m
IoU=0.253
failure=success



Wrong detection
True-name query ablation: PickClutterYCB
GroundingDINO / Crop-median depth
none L0 seed 124
success=False error=0.890 m
IoU=0.021
failure=wrong detection selected



No detection
True-name query ablation: PickClutterYCB
GroundingDINO / Crop-median depth
none L0 seed 111
success=False error=n/a
failure=no detection



Depth invalid
Detector-bridge sweep: PickClutterYCB
First detection / Box-center depth
partial target occlusion L3 seed 20
success=False error=n/a
failure=depth invalid



Large target error
True-name query ablation: PickClutterYCB
GroundingDINO / Crop-trimmed median
none L0 seed 124
success=False error=0.890 m
IoU=0.021
failure=wrong detection selected



Low IoU but 3D success
Detector-bridge sweep: PickCube
GroundingDINO / Box-center depth
partial target occlusion L2 seed 4
success=True error=0.050 m
IoU=0.016
failure=success